

Embedded Systems Engineer – Sensor System (BLE + Signal Processing)

Posted 3 hours ago



Worldwide

Summary

This is a validation-driven MVP project, focused on proving sensing quality and system feasibility before moving into a compact integrated design.

Scope (Phase 1)

- Dev board-based prototyping (nRF52 / ESP32 or similar)
- Sensor integration (vibration + potential acoustic emission)
- Data acquisition and basic signal processing (RMS, FFT, feature extraction)
- BLE communication to external gateway
- Battery-powered system design with attention to efficiency

What we're looking for

- Strong embedded systems background (hardware + firmware)
- Experience with BLE and low-power design
- Understanding of signal acquisition in noisy/real-world environments
- Ability to think at system level (not just coding or PCB design)

Project Approach

This is not a fixed-spec build. We are looking for someone who can:

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About the client

Payment method verified

Phone number verified

★★★★★ 0

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United States

Queens County 3:44 PM

5 jobs posted

40% hire rate, 2 open jobs

\$977 total spent

4 hires, 4 active

\$17.53 /hr avg hourly rate paid

51 hours

Member since Mar 20, 2026

Job link

<https://www.upwork.com/jobs/>

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- Help validate assumptions
- Identify risks early
- Contribute to architecture decisions

There is a clear path toward custom hardware and mechanical integration after MVP validation.

More than
30 hrs/week
Hourly

3-6 months
Duration

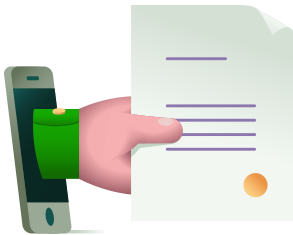
Intermediate
Experience
Level

\$25.00 - \$30.00
Hourly

Contract-to-hire opportunity

This lets talent know that this job could become full time.

[Learn more](#)



Project Type: Ongoing project

Skills and Expertise

Mandatory skills

Embedded C

Real-Time Operating System

AVR Microcontroller

Embedded Systems

Microcontroller Programming

Sensor Integration

ARM Microcontroller

Preferred qualifications

Activity on this job

Proposals: 

20 to 50

Last viewed by client:

 1 hour ago

Interviewing: 1

Invites sent: 5

Unanswered invites: 4

Bid range - High \$85.00 | Avg \$33.38 | Low \$20.00

Client's recent history (4)

Jobs in progress

[IoT and Embedded Engineer Needed to Build a Working Prototype](#)

No feedback given

To freelancer: [Bakmeedeniye gedara praveena s.](#) No feedback given

Apr 2026 - Apr 2026

12 hrs @ \$5.00/hr

Billed: \$71.50

Other open jobs by this Client (1)

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